Forecasting Three-Day High-Flow Exceedance at Gauges Unseen in Space and Time: A Leakage-Controlled Benchmark on CAMELS-IND

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***Abstract*—Flood warnings in India are issued gauge by gauge, yet machine-learning flood forecasters are often scored on random row splits that let neighbouring days of the same river leak between training and test data. We build a leakage-controlled benchmark on CAMELS-IND for one operational question: will a gauge's flow reach its 95th-percentile level, a high-flow proxy for a flood level exceeded on about 18 days a year, within the next three days? We train on 193 gauges for 1980–2008, select models and thresholds on 2009–2014, and test on 48 unseen gauges in 2015–2020. A gradient-boosted model reached a balanced accuracy of 0.898 (95% gauge-bootstrap interval 0.886–0.910) and a PR-AUC of 0.737, above persistence and gauge climatology, and not significantly different from the dataset's regional LSTM simulation (0.873), which is driven by observed future rain and was trained on years that overlap ours in 2015. On days when the river was still below the level, where a warning matters, balanced accuracy was 0.857 and recall fell from 90.3% one day ahead to 80.0% three days ahead (96.7% to 89.5% over all flood days), again not significantly different from the LSTM. A refit scored on a random row split reached 0.915, 1.7 points higher on its own test set. Five other gauge splits gave 0.877–0.898 and leave-basin-out folds 0.866–0.906. Precision was 0.328, with most false alarms in the monsoon, so a positive forecast should trigger review by an officer.**

***Index Terms*—CAMELS-IND, data leakage, flood forecasting, gradient boosting, hydrological benchmark**

I. Introduction

Indian flood warnings are issued per gauge, and an emergency officer needs to know whether the flow is likely to reach its flood level in the next few days. Models trained on many catchments now simulate streamflow well, even in ungauged basins [1]–[3]. Reported skill, however, depends on how a model is evaluated. When rows are split at random, days from the same flood fall on both sides of the split, and part of the test score then rewards memory of the event [4], [5].

For India, CAMELS-IND provides daily forcing, observed streamflow and catchment attributes for peninsular Indian catchments [6], in the format of the CAMELS family [7], [8]. It also ships a regional LSTM streamflow simulation, a strong deep-learning reference. No short-range flood-exceedance benchmark on these gauges yet controls leakage, compares against simple and strong references, and reports the numbers an operations room needs: alarm rates, onset lead time and calibration.

Our contributions are:

1) A leakage-controlled task and split: gauges held out in space and time, flood thresholds from training years only, model and threshold selection on validation only, and one scoring of the selected model on the test set; later test-set diagnostics are marked post hoc (Fig. 1).
2) A measurement of how much the evaluation protocol changes the score: one model configuration scored under random, temporal, spatial and space-and-time splits.
3) A comparison with persistence, the gauge's own climatology and the CAMELS-IND regional LSTM simulation, with gauge-cluster bootstrap intervals.
4) A feature-group ablation that shows what the forecast relies on, including a variant that needs no streamflow reading at forecast time.
5) An operational analysis of operating points, calibration, per-gauge skill and the lead time of warnings before flood onsets.

II. Related Work

Large-sample learning. The CAMELS datasets standardised forcing, flow and attributes across hundreds of basins [7], [8]. LSTMs trained on them learn regional behaviour [1], predict ungauged basins [2] and simulate extremes better than conceptual models [9]; at global scale, a learned model forecasts extreme floods in ungauged watersheds with skill comparable to an operational system [3]. Graph models that encode the river network have so far not beaten models that ignore it [10]. Most of this work simulates streamflow; flood *exceedance*, the quantity a warning is issued on, is usually derived afterwards.

Operational forecasting. Physically based ensemble systems such as GloFAS [11] dominate operational flood forecasting [12]. Machine learning has also reached operations: a system run for India and Bangladesh combines learned river-stage forecasts with inundation models [13]. In India, the INDOFLOODS catalogue documents historical events [14] and many studies apply tree ensembles or neural networks to single basins [15], often with random splits.

Evaluation. Leakage through the split is a leading cause of irreproducible machine-learning results [4], and data with spatial or temporal structure need blocked validation [5]. For rare events, precision–recall curves are more informative than ROC curves [16], and probabilistic forecasts should be checked for calibration [17].

To our knowledge, no study has benchmarked short-range flood-exceedance forecasting on CAMELS-IND with a space-and-time held-out split against both a learned streamflow simulation and simple references.

III. Data and Task

A. Target

For gauge g on day t the target is

$$y_{g,t} = \mathbb{1}\Big[\max_{h\in\{1,2,3\}} Q_{g,t+h} \;\geq\; q_g^{95}\Big], \tag{1}$$

where Q is observed streamflow and q_g^{95} is the 95th percentile of the gauge's flow over the training years 1980–2008 only. A day with $y_{g,t} = 1$ is a *flood window*. The level is a statistical proxy, exceeded on about 18 days a year, not an official danger level, which CAMELS-IND does not provide; a stricter 99th-percentile level is tested in Section V-D. A three-day horizon matches the time an officer needs to stage teams. Days without a flow observation in the horizon are dropped: 281,360 gauge-days (9.3% of the observed spans). They were not concentrated in floods: 26% fell in the monsoon against 34% of the kept days, and 1.5% followed a day at or above q_g^{95} against 4.9%.

B. Inputs

Table I lists the 50 inputs, all known at the end of day t. Flow features are scaled by q_g^{95}, so they read as "how close the river is to its flood level". Hydrological signatures of CAMELS-IND are excluded, because they are computed over the full 1980–2020 record and would leak test-period information; future rainfall is excluded as well. Rainfall is IMD's gridded product and soil moisture, humidity and evapotranspiration come from the IMDAA reanalysis and GLEAM [6]; these are available with a delay, so in real time they would be replaced by forecasts or near-real-time estimates, and the skill reported here is an upper bound for that setting.

TABLE I
Input groups (all known at the end of day t)

Group	Content	Inputs
Streamflow	lags 0–3, changes, 7/30-day means and maxima	11
Rainfall	lags 0–3, 3–90-day sums, 7-day maximum	10
Weather	temperature, humidity, evapotranspiration	5
Soil moisture	four layers and recent changes	6
Season	day of year (sine, cosine)	2
Catchment	area, slope, aridity, land cover, soils	16

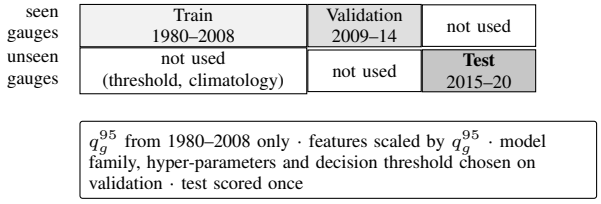


Fig. 1. Leakage-controlled split. The primary test uses gauges and years that played no part in training, model selection or threshold choice. The unseen gauges' 1980–2008 record is used only to set their flood threshold and climatology.

C. Split

Fig. 1 shows the split. All 242 gauges in the streamflow file have forcing data and at least three years of flow in 1980–2008. The data description counts 228 catchments with more than 30% flow availability [6], but all 242 gauges of the version we use pass that filter, so none is excluded. A group split puts 20% of gauges aside (seed 42), so the primary test asks about rivers the model has never seen, in years it has never seen; one of the 49 unseen gauges has no flow record in 2015–2020, which leaves 48 test gauges. Training uses the other gauges in 1980–2008 (1,641,197 gauge-days), validation the same gauges in 2009–2014 (349,645), and the test the unseen gauges in 2015–2020, all labelled by (1) (52,994 gauge-days, of which 5.5% are flood windows).

IV. Methods

Model. Six candidates (logistic regression, random forest, histogram gradient boosting and XGBoost [18] at three depths) and a refit of the depth-6 model were compared on validation balanced accuracy (Table II), and the decision threshold of the selected depth-6 XGBoost model, 0.45, was set on validation as well. It uses a learning rate of 0.05, subsampling of rows and columns at 0.8, a minimum child weight of 20, L2 regularisation of 5, a positive-class weight equal to the ratio of negative to positive training rows, and early stopping on validation PR-AUC, and it was refit on all training rows. Logistic regression reached a validation balanced accuracy of 0.899 against 0.911 for the selected model, so the selected

TABLE II
CANDIDATE MODELS ON VALIDATION (SEEN GAUGES, 2009–2014)

Model	BA	PR-AUC
Logistic regression	0.899	0.751
Random forest	0.905	0.781
Hist. gradient boosting	0.909	0.791
XGBoost, depth 4	0.909	0.790
XGBoost, depth 6	0.909	0.792
XGBoost, depth 8	0.909	0.793
XGBoost, depth 6, selected refit	0.911	0.796

refit adds little balanced accuracy but 4.5 points of PR-AUC. With TPR and TNR the true positive and true negative rates,

$$\mathrm{BA} = \tfrac{1}{2}\left(\mathrm{TPR} + \mathrm{TNR}\right). \tag{2}$$

References. Persistence forecasts a flood window if the river is above its flood level today. A *trend rule* needs no learning: its score is $Q_{g,t}/q_g^{95} + \max(0, Q_{g,t} - Q_{g,t-3})/q_g^{95}$, so a high or rising river raises an alarm. *Climatology* forecasts the gauge's own 1980–2008 rate of flood windows within ± 15 days of the calendar day. The *LSTM simulation* is the regional LSTM streamflow of CAMELS-IND [6], scored as $\max_{h \le 3} \hat{Q}_{g,t+h}/q_g^{95}$. It is driven by observed forcing, so it sees the rain of days $t{+}1$ to $t{+}3$, which no real forecast has. It was trained by the dataset authors on 159 catchments over 1991–2015 and tested on 2016–2020 [6], so it has most likely seen our test gauges and, in 2015, one of our test years; it is an optimistic reference, not a held-out competitor. Thresholds of the references were set on validation like the model's.

Analyses. We refit the selected configuration on the same 500,000-row training sample with one input group removed at a time, and with the streamflow and season inputs only. Confidence intervals come from 1,000 resamples of the test gauges, which keeps each river's days together. Calibration is the expected calibration error over ten bins, before and after isotonic regression fitted on validation. A flood onset is a day on which the river reaches q_g^{95} after at least seven days below it; a method warns of it if it forecasts a flood window one, two or three days earlier. For lead-day and seasonal analyses, a flood day at lead h is a day t with $Q_{g,t+h} \ge q_g^{95}$, and the monsoon is June to September. Sensitivity refits (Section V-D) use the ablation's 500,000-row sample: five more random gauge splits (seeds 1 to 5), a leave-basin-out evaluation in which the CAMELS-IND river basins are grouped into five folds and every basin is held out once, and the same task with the 99th percentile as level.

Reproducibility. The primary split and the bootstrap use seed 42, and the 49 held-out gauges are stored with the model metadata; code, model and split lists will be released after review. Re-scoring the stored model reproduces its recorded test balanced accuracy exactly, and every number in this paper is generated from the experiment outputs by a script that also checks each claim made about it.

TABLE III
THREE-DAY EXCEEDANCE ON 48 UNSEEN GAUGES, 2015–2020

	all rows					onset rows	
Method	BA	95% CI	PR-AUC	Rec.	Prec.	BA	PR-AUC
XGBoost (ours)	0.898	0.886–0.910	0.737	0.902	0.328	0.857	0.369
No-flow XGB.	0.853	0.820–0.880	0.494	0.865	0.239	0.846	0.257
LSTM sim.†	0.873	0.829–0.911	0.589	0.826	0.376	0.877	0.421
Trend rule	0.828	–	0.605	0.754	0.305	0.734	0.158
Climatology	0.802	0.774–0.829	0.203	0.891	0.152	0.802	0.102
Persistence	0.750	0.724–0.778	0.428	0.508	0.789	0.500	0.026

CI: gauge-cluster bootstrap. Onset rows: river below q_g^{95} on day t.
†Driven by observed future rain; covers every test row.

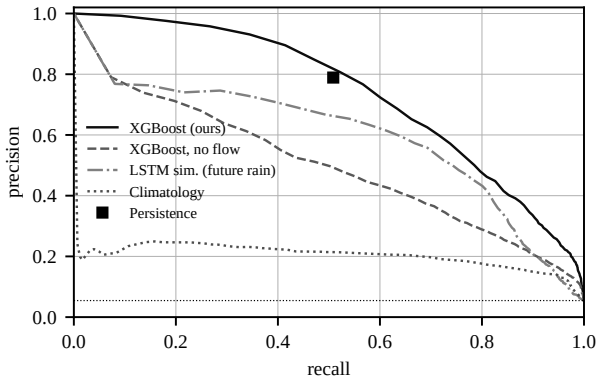


Fig. 2. Precision–recall curves on the unseen test gauges. The square is persistence; the dotted line is the flood rate.

V. RESULTS

A. Skill on Unseen Gauges

Table III and Fig. 2 give the main comparison, with balanced accuracy as defined in (2). The model scored above persistence and climatology in balanced accuracy and PR-AUC, and its bootstrap interval for the gain in balanced accuracy over persistence (0.123–0.175) excludes zero. Persistence was more precise (0.789 against 0.328) but misses every flood that has not yet begun. The model's recall of 0.902 means that it found nine in ten flood windows at rivers it had never seen. Many of these windows are easy: a river above its level usually stays there. On onset rows, where the river is still below the level, the model's balanced accuracy was 0.857 and its PR-AUC 0.369, against 0.734 and 0.158 for the trend rule, 0.802 and 0.102 for climatology and 0.500 for persistence, which cannot anticipate an onset.

The LSTM simulation is a demanding reference because it knows the rain that is still to fall. Over all rows its balanced accuracy was 0.873 against 0.898 for the model, and the paired gauge-bootstrap interval of the difference, -0.009 to 0.063, includes zero. The two also run at different operating points: at the LSTM's alarm rate (120 per 1,000 gauge-days) the model reached a precision of 0.394 and a recall of 0.864, against 0.376 and 0.826. On onset rows the LSTM scored higher (0.877 against 0.857), as its access to future rain would suggest, but the difference interval (-0.045 to 0.011) includes zero.

TABLE IV
SAME CONFIGURATION, FOUR PROTOCOLS (500,000-ROW REFIT)

Test set	Gauges	Flood %	BA	95% CI	PR-AUC	Prec.
Random rows	242	7.4	0.915	0.911–0.919	0.810	0.426
New years	181	6.1	0.892	0.885–0.898	0.724	0.349
New gauges	49	7.6	0.905	0.898–0.912	0.789	0.424
New gauges and years	48	5.5	0.898	0.886–0.911	0.736	0.340

Random: 70/10/20% of gauge-days; others trained on seen gauges, 1980–2008.
New years: seen gauges, 2015–20. New gauges: unseen, 1980–2008.

TABLE V
FEATURE-GROUP ABLATION (SAME 500K TRAINING SAMPLE)

Inputs	n	BA	PR-AUC	Rec.	Onset BA
All inputs	50	0.898	0.736	0.897	0.852
No streamflow	39	0.853	0.494	0.865	0.846
No catchment	34	0.895	0.734	0.907	0.855
No soil moisture	44	0.896	0.732	0.895	0.854
No rainfall	40	0.897	0.730	0.901	0.857
Streamflow + season	13	0.875	0.685	0.882	0.810

Onset BA: rows where the river is below its flood level on day t.

TABLE VI
SENSITIVITY REFITS (500,000-ROW TRAINING SAMPLE)

Evaluation	Test gauges	BA	PR-AUC	Onset BA
5 gauge splits	range	0.877–0.898	0.706–0.764	0.822–0.851
5 basin folds	range	0.866–0.906	0.648–0.829	0.785–0.864
99th pct. level	48	0.853	0.364	0.798
met. + season only	48	0.832	0.447	0.825
no class weight	48	0.902	0.737	0.863

B. Effect of the Evaluation Protocol

To measure what the split itself does to the score, we trained the same configuration on a 500,000-row sample under four protocols, each with its own validation set for early stopping and threshold, and scored it once (Table IV). The random row split gave the highest balanced accuracy, 0.915, against 0.898 on unseen gauges in later years; the two bootstrap intervals (0.9115–0.9189 and 0.8862–0.9111) only just separate, and they come from different test sets, so the gap is indicative rather than tested. The random split scored 1.7 points higher in balanced accuracy and 7.4 points higher in PR-AUC. PR-AUC ranged from 0.724 (new years) to 0.810 (random rows). Holding out later years lowered PR-AUC more than holding out gauges (0.724 against 0.789), although these are different test sets with different flood rates. In the spatial protocol, the test years are also the years from which each gauge's flood threshold was computed. Flood prevalence differs between protocols (7.4% of random rows against 5.5% in the held-out test), which moves PR-AUC on its own.

C. What the Forecast Relies On

Table V shows the ablation. Over all rows, removing the streamflow inputs cost the most, lowering balanced accuracy from 0.898 to 0.853 and PR-AUC from 0.736 to 0.494, while removing soil moisture, rainfall or catchment attributes cost little. Much of that score comes from rivers that are already high and stay high. On onset rows, where the river is still below its flood level, removing streamflow lowered onset balanced accuracy only from 0.852 to 0.846, streamflow and season alone reached 0.810 and meteorology and season alone 0.825. The two kinds of input are complementary for onsets, and neither alone reaches the full model. Within the meteorological inputs the groups are redundant: removing rainfall, soil moisture or catchment attributes one at a time changed onset balanced accuracy by at most 0.0051 (each removal raised it slightly), so these single-group results cannot rank them. The no-flow variant also scored above climatology overall (0.853 against 0.802, with overlapping intervals), so it is a possible fallback when a gauge reading is missing.

The selected model's gain importance agrees: streamflow inputs account for 67.5% of the total gain, led by today's flow relative to the flood level (40.5%) and its seven-day maximum (23.9%), followed by rainfall (13.6%, of which today's rain alone 8.5%). Gain importance favours inputs used in many splits, so we use it only as a check on the ablation.

D. Sensitivity to Split and Level

Table VI repeats the evaluation under other splits and a stricter level. Over five more random gauge splits the balanced accuracy ranged from 0.877 to 0.898 and the onset balanced accuracy from 0.822 to 0.851, so the primary refit (0.898) lies at the top of that range. Holding out whole river basins gave 0.866–0.906 (onset 0.785–0.864), below every random split in 2 folds, but persistence also varied across folds (0.699–0.812): the gain over persistence was 0.119–0.136 over the random splits and 0.094–0.186 over the basin folds. With the 99th percentile as level (1.6% of test rows positive) the balanced accuracy was 0.853 and the PR-AUC 0.364 at a precision of 0.117. Skill per test year ranged from 0.873 (2017) to 0.914 (2016) over 2015–2018; 2019 and 2020 have only 2,102 gauge-days in the dataset. Gauges with dams upstream scored higher than those without (median per-gauge balanced accuracy 0.911 for 37 gauges against 0.875 for 9; Spearman $\rho = 0.37$ with the number of dams, $p = 0.02$, but $\rho = 0.20$, $p = 0.23$ with the reservoir index); these attribute tests were not corrected for multiple comparisons. The level was also exceeded less often in the test years (3.9% of days) than in the training years that defined it (5.0%).

E. Operational Behaviour

Precision is the main weakness. At the validation threshold the model raised 150.2 alarms per 1,000 gauge-days at a precision of 0.328 (Table VII); raising the threshold to 0.75 halved the alarms and lifted precision to 0.539, at a recall of 0.760. Because a positive forecast in our system leads to human review rather than automatic action, the validation threshold is a reasonable default. The raw probabilities were over-confident (ECE 0.102, gauge-bootstrap interval 0.094–0.110; Brier score 0.065, worse than climatology's 0.048); isotonic recalibration on validation data lowered the ECE to 0.005 (0.002–0.008) and the Brier score to 0.025. Raw ECE

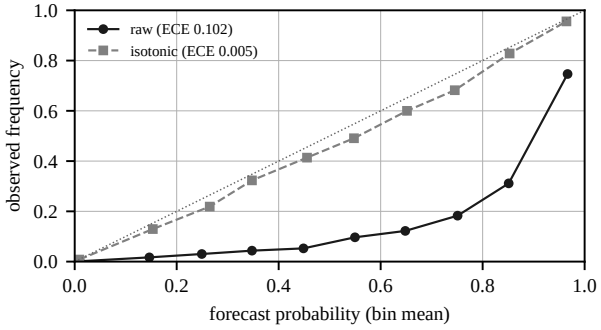


Fig. 3. Reliability of the forecast probabilities on the unseen test gauges, before and after isotonic recalibration fitted on validation data (bins with at least 50 gauge-days).

TABLE VII
OPERATING POINTS ON THE TEST SET (THRESHOLD 0.45 SET ON VALIDATION)

Threshold	Recall	Precision	Alarms / 1000 days
0.30	0.942	0.261	197.2
0.45	0.902	0.328	150.2
0.60	0.845	0.413	111.6
0.75	0.760	0.539	77.0

was 0.250 in June to September and 0.026 in the other months. In the highest well-populated bin of Fig. 3 the raw forecast averaged 0.97 while floods followed in 0.75 of cases. A refit without the positive-class weight, run after the test results were known, had a raw ECE of 0.008 at a balanced accuracy of 0.902 (onset 0.863), against 0.898 and 0.852 for the same sample with the weight. The weighting therefore caused most of the over-confidence and bought no skill; the monsoon gap in ECE reflects where the high scores fall. The alarms at the validation threshold are computed from the raw score and are unaffected, but a raw score must not be shown to an officer as a probability; the recalibrated one can be.

Table VIII gives the lead of the warnings. The model warned three days ahead for 63.9% of the 266 onsets at 44 gauges, with a mean lead of 2.58 days when it warned. Climatology (89.1%) and the LSTM simulation (78.9%) appear better on this measure, but for different reasons: climatology raises an alarm on most monsoon days, which its precision of 0.152 reflects, and the LSTM simulation is given the rain that is still to fall. Lead counts alone therefore reward methods that alarm constantly: random alarms at the model's average rate would warn of 38.6% of onsets within three days, and at its monsoon rate of 75.2%. Officers act on episodes rather than days; grouped into runs of consecutive alarm days, the model raised 1,441 alarm episodes at the test gauges, of which 1,106 (77%) contained no flood window.

Skill was consistent across rivers: over the 46 test gauges with at least one flood window, the median per-gauge balanced accuracy was 0.900 (interquartile range 0.860–0.926), and the worst gauge reached 0.769. For the 266 flood onsets at the test gauges, the model warned one to three days ahead in 91.4%

TABLE VIII
WARNING LEAD BEFORE THE 266 FLOOD ONSETS (% OF ONSETS)

Method	3 days	2 days	1 day	none	Mean lead
XGBoost (ours)	63.9	16.5	10.9	8.6	2.58
No-flow XGB.	72.2	14.7	7.1	6.0	2.69
LSTM sim.†	78.9	4.5	0.0	16.5	2.95
Climatology	89.1	0.4	0.0	10.5	3.00

Mean lead in days, over warned onsets. †Driven by observed future rain.

TABLE IX
SKILL BY SEASON ON THE UNSEEN TEST GAUGES

Months	Gauge-days	Flood %	BA	Rec.	Prec.	Alarms/1000
Jun–Sep	17,992	12.7	0.811	0.915	0.311	372
Oct–May	35,002	1.8	0.918	0.857	0.415	36

of cases, against 83.5% for the LSTM simulation, 89.5% for climatology and 0.0% for persistence, which by definition cannot anticipate an onset. The no-flow variant warned of 94.0%, more than the full model, but at a lower precision (0.239), and onset counts reward frequent alarms.

F. Where the Forecast Succeeds and Fails

Lead day. The target pools three lead days, so we split the recall by the day on which the river is above its flood level. The model raised an alarm on day t for 96.7% of flood days at $t{+}1$, 93.3% at $t{+}2$ and 89.5% at $t{+}3$. On rivers still below their flood level on day t, which is where a warning is useful, the rates were lower: 90.3%, 85.2% and 80.0%.

Season. Table IX separates the monsoon months. In June to September the model raised 372 alarms per 1,000 gauge-days at a precision of 0.311, against 36 and 0.415 in the rest of the year, and 86% of all false alarms fell in the monsoon. Most of these were not near misses: in the median monsoon false alarm the river peaked at 41% of its flood level over the next three days, and only 21% of them reached 70–100%. The model thus responds to monsoon conditions in general more than to an approaching flood.

Catchment. Over the 46 test gauges with flood windows, per-gauge balanced accuracy rose with catchment area (Spearman $\rho = 0.44$, $p = 0.002$), which is consistent with large rivers rising slowly enough for today's flow to describe the next three days. Aridity, mean rainfall and slope showed no relation ($p \geq 0.64$). We expect, but have not tested, that rainfall forecasts would help most in small, fast catchments.

VI. DISCUSSION AND LIMITATIONS

First, the inputs play different roles. Today's flow carries the overall score, while at flood onsets neither input group alone matched the full model, so an operations room needs both a reliable gauge feed and good meteorological inputs; when the feed fails, the no-flow variant scored above climatology. Second, precision is the weak point: most false alarms fell in June to September on rivers that stayed well below their flood level. The LSTM simulation, which is given the observed future rain, reached a precision of only 0.376, so better rainfall

information alone may not fix it. Until precision improves, an alarm should start a review, not an evacuation. Third, a refit scored on a random row split reached a balanced accuracy 1.7 points higher than the held-out protocol (Table IV). The test sets and flood rates differ, so not all of this gap is leakage, but scores from random row splits should not be read as skill at a new gauge, and published Indian flood models can only be compared under the same split.

In deployment the full model would run while the gauge feed is live and the no-flow model when it is not, with recalibrated scores and officer review of every alarm; the fallback was evaluated offline only.

For studies that report flood forecasts on Indian rivers, the results suggest a minimal protocol: hold out both gauges and later years and report the split with the score (Table IV); set flood thresholds, models and decision thresholds from training and validation years only; report precision and alarm rates by season (Table IX) and skill on onset rows (Table V); and check calibration before showing a score as a probability (Fig. 3).

The study has limits. The CAMELS-IND gauges are in peninsular India, so skill may not transfer to Himalayan or Brahmaputra rivers. The 95th-percentile threshold is a statistical proxy for an official danger level. The LSTM reference was trained by the dataset authors on years and, most likely, gauges that overlap our test set, and it sees future rain, so the comparison with it is indicative only. We pool three lead days into one target. Intervals resample gauges but not years, and gauges in one monsoon share storms, so they are likely too narrow; skill per full test year ranged from 0.873 to 0.914, and was lower in the partial years 2019 and 2020 (0.783 and 0.767). The inputs other than rainfall come from reanalysis, so real-time skill will be lower. The ablation and protocol models were refitted on a 500,000-row sample, so they are compared with each other; the all-input refit scored 0.898, as the selected model trained on all rows did (0.898).

VII. Conclusion

On CAMELS-IND gauges held out in space and time, a gradient-boosted model forecast three-day exceedance of the 95th-percentile flow with a balanced accuracy of 0.898 and a PR-AUC of 0.737, above persistence and the gauge's own climatology, and comparable to the dataset's LSTM simulation, which sees future rain. Skill above persistence held across other gauge splits and whole held-out basins. On flood onsets the skill was lower (0.857), not significantly different from the LSTM, and neither streamflow nor meteorological inputs alone matched the full model. Precision of 0.328, lowest in the monsoon, and miscalibrated raw scores make the model a trigger for review by an officer, not an order. A random row split scored 1.7 points higher in balanced accuracy on its own test set. Code and split lists will be released after review.

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